Preliminary Ecological Appraisal

Date prepared:	07/05/2025
Site name:	124 Cheltenham Road
Site address:	Bishops Cleeve, Cheltenham, Gloucestershire, GL52 8LZ
Grid reference:	SO95512675
Local authority:	Tewkesbury Borough Council
Client:	Boughton Butler
Survey dates:	01/10/2025
Prepared by:	Phillipa Stirling MSc ACIEEM
Reviewed by:	Will Prestwood BSc

Contents

1 INTRODUCTION

1.1 BACKGROUND TO DEVELOPMENT

Planning permission will be sought for the construction of a dwelling in the garden of 124 Cheltenham Road.

Arbor Vitae were commissioned by Boughton Butler to undertake a Preliminary Ecological Appraisal in order to assess the impact of the development on habitats and protected species.

The application will be for a self-build dwelling and therefore is exempt from Biodiversity Net Gain.

1.2 SCOPE OF SURVEY

The survey is primarily designed to:

- Identify and record habitats and important ecological features on site;
- Evaluate the potential of the proposed development site to provide opportunities for protected species;
- Determine any likely impact which the development and landscape proposals may have on these.
- Identify opportunities for the enhancement of habitats and biodiversity features on site.

1.3 KEY PRINCIPLES

All ecological surveys conducted by Arbor Vitae Environment Ltd are underpinned by the following key principles, as outlined by CIEEM (2018):

Avoidance - Seek options that avoid harm to ecological features (for example, by locating on an alternative site).

Mitigation - Adverse effects should be avoided or minimized through mitigation measures, either through the design of the project or subsequent measures that can be guaranteed – for example, through a condition or planning obligation.

Compensation - Where there are significant residual adverse ecological effects despite the mitigation proposed, these should be offset by appropriate compensatory measures.

Enhancements - Seek to provide net benefits for biodiversity over and above requirements for avoidance, mitigation or compensation.

2 SITE DESCRIPTION

2.1 LOCATION, LANDSCAPE, AND BACKGROUND

Cheltenham Road forms a main street through Bishops Cleeve, Tewskesbury just off the A435. The area of Bishops Cleeve sits to the north of Cheltenham in Gloucestershire. The area is predominantly made up of residential properties with their associated outdoor space, driveways and road systems.

The plans for the site will include the installation of a new access drive along the north boundary of the property, provision of parking spaces, and a new dwelling.

3 SURVEY METHODOLOGY

3.1 DESK STUDY

An initial desk study was composed to gain background information regarding any protected species or designations within the area. The main sources of information were MagicMap, and NBN Atlas.

3.2 SITE SURVEY

A site visit was made on 31/07/2025. The survey was carried out in accordance with CIEEM (2017) best practice guidelines. The objective of the survey was to find and record any signs of use by protected species and to note the habitat features present.

An assessment of the available habitats both on and adjacent to the site led to consideration of the potential of the site for the following protected species:

- Bats
- Breeding birds
- Great Crested Newt
- Hedgehog

The survey methodology was tailored to evaluate the area for these species in the following ways:

Bats

The site was assessed in terms of its suitability to support bat species. Hedgerow habitat and nearby potential habitat were assessed and recorded and potential impacts from the proposals considered.

Breeding birds

The site was assessed in terms of its suitability to support breeding bird populations. Hedgerow habitat and nearby potential habitat were assessed and recorded.

Great crested newt

A desk study and a ground search were conducted to search for any areas of open water within 250 metres. Waterbodies were then assessed based on the Habitat Suitability Index for great crested newts (Oldham et al., 2000 and ARG UK, 2010).

Hedgehog

The site was assessed in terms of its suitability for hedgehog e.g. areas of rough vegetation, built structures with cavities beneath, availability of feeding areas.

3.3 PERSONNEL

The survey was carried out by Phillipa Stirling MSc ACIEEM: Ecologist.

Natural England bat Level 1 licence number: 2021-52205-CLS-CLS, GCN Level 1 licence number: 2019-42631-CLS-CLS, hazel dormouse Level 1 licence number: 2023-11248-CL10A-DOR.

3.4 CONSTRAINTS

There were no constraints to the survey work being completed.

4 SURVEY RESULTS

4.1 DESK STUDY

No designated sites were found within 1km of the survey area. The search included Ramsar, SSSI, SAC, SPA, LWS, NNR and LNR. [1]

Results from the desk study revealed that within a 1km radius of the proposed development site the following protected species have been recorded:

Species	Distance	Protection
	Mammals	
Hedgehog	0.1km	s.41 NERC
Badger	0.3km	Protection of Badgers Act 1992, Wildlife and Countryside Act 1981.
Common pipistrelle	0.9km	European Protected Species, Wildlife and Countryside Act 1981.
	Birds	
Kingfisher Merlin Peregrine Kestrel Brambling Redwing Fieldfare Barn owl	0.3-1km	Wildlife and Countryside Act 1981.
	Reptiles	
Slow worm	0.4km	Wildlife and Countryside Act 1981.
	Amphibians	
Great crested newt	0.7km	European Protected Species, Wildlife and Countryside Act 1981.

[1] **SSSI**: Site of Special Scientific Interest, **SAC**: Special Area of Conservation, **SPA**: Special Protection Area, **LWS:** Local Wildlife Site **NNR:** National Nature Reserve, **LNR**: Local Nature Reserve.

4.2 HABITATS ON SITE

All habitats are classified using JNCC's Phase 1 Habitat Survey Handbook (JNCC, 2010).

Habitat type & description	Predicted Impact	Proposed mitigation measures
Artificial unvegetated The existing driveway and a section of path is gravel.	It is likely that these areas will be re-surfaced with tarmac, or similar. This will have little ecological impact.	None required.
Introduced shrub There are a number of ornamental shrubs which have been planted within the garden. They are mostly mature and non-native, such as laurel, *Leylandii, Cotoneaster,* rhododendron, forsythia.	Some of these shrubs will be removed to facilitate the new driveway for the property. They have little ecological value, and the loss will have no wider implications.	None required.
Vegetated garden Areas of well-mown lawn are found through much of the site. Typical species such as ryegrass, creeping bent, creeping red fescue, white clover, dandelion, self-heal, creeping buttercup, and lawn mosses were identified.	Around 380m2 of lawn will be lost in order to accommodate the new dwelling on site. This habitat will be partly re-instated upon completion of the dwelling, and its temporary loss is of very little ecological consequence.	Areas of lawn will be re-instated once the dwelling is complete which will roughly equate to the amount being lost.
Allotment A small vegetable garden with fruits and root vegetables. There is also a small greenhouse on site.	The plans will result in the total loss of this habitat, and it is unlikely to be re-instated. Whilst classed as an 'urban' habitat, the allotment is likely to provide habitat for invertebrates, birds, and small mammals.	A wildlife friendly landscaping scheme should be adopted throughout the new garden area, to include herbaceous borders and shrubs. At least two Woodcrete invertebrate shelter will be installed at the base of the boundary hedgerow.
Developed land There ar**e** a shed and a garden room located in the area for the new dwelling.	These structures are of no ecological significance and will be removed to facilitate the development.	None required.

Non-native hedge H1-H4 are all non-native garden hedges consisting of *Leylandii, Cotoneaster,* laurel, and privet.	Part of H2 and H4 (Figure 3) will be removed to facilitate the plans on-site. The loss of these non-native hedges will be of limited consequence in terms of habitats. These features are likely to form microclimates for invertebrates and small animals, and some replacement hedge planting is recommended.	It is recommended that a new hedge section is planted along the proposed access track and east boundary of the car port, in order to replace hedge losses on site. Hedgerow planting should ideally be native, although replacing like for like in this instance is acceptable.
Native hedge H5 is a short length of native hedge which includes hawthorn, wild privet, elder, and dogrose.	Plans will not result in the loss or damage of this hedge.	None required.
Individual trees T1: ornamental maple, small (outside red line). T2: *Leylandii,* small. T3: *Leylandii,* small (outside red line). G1: a group of small *Leylandii* which are small (outside red line). G2: a group of small apple trees, small. G3: a group of small *Leylandii* which are small (outside red line).	T2, T3, G1, and G2 will need to be removed to facilitate the construction of the property. G1, T2 and T3 are non-native trees, and their loss will be inconsequential. The group of small apple trees is not of particularly high ecological significance although they do provide a small ecological niche.	Ideally, plans for the site should include the provision of a small group of fruit trees to replace those being lost. A minimum of 5 native fruit trees should be planted in a corner of the garden to provide a small orchard area. Apple, pear, plum, and cherry are all suitable for this purpose.
Ornamental pond There is a small ornamental fishpond outside of the red line boundary. The pond is around 1m2 and is fully lined with a butyl liner. The edges of the pond are steep, and there are coping stones around the edge.	The pond will not be impacted by the proposals.	None required.

4.3 PROTECTED SPECIES

Protected species	Predicted Impact	Proposed mitigation measures
Bats	None of the temporary structures in the garden provide suitable roosting opportunities for bat species. None of the trees which are to be removed provide suitable roosting features for bat species either. The hedgerows around the site provide good vegetative corridors and these, along with most of the existing shrubs/vegetation will be retained as part of the plans. Any impact to bat species may arise from the installation of new artificial lighting at the site. Mitigation will be required.	A Wildlife Sensitive Lighting Plan will be adopted as part of plans for the site. This will focus on protecting the boundary hedgerows from light spill. Appendix 2.
Breeding birds	Shrubs, trees, and hedgerows on site provide suitable nesting opportunities for generalist and garden birds. The loss of some areas of shrub and trees has the potential to disturb breeding birds, if done during the breeding season. Mitigation will be required.	Appendix 4.
Great crested newt	There is a single, very small ornamental pond within the garden plot of 124 Cheltenham Road. The pond provides 'poor' suitability as a GCN breeding pond, according to the Habitat Suitability Index. See Appendix 5 for results. The nearest record of GCN is 700m away and there are no other ponds within 250m of the site. The plans are unlikely to have any impact upon GCN, and no further survey work is required.	None required.
Hedgehog	Hedgehog has been recorded locally and the habitats on site provide suitable terrestrial habitat opportunities for hedgehog. Any new boundary fences and walls will present solid barriers to wildlife, particularly hedgehogs. This is important because a third of hedgehogs have been lost in the last 20 years, and one major cause is barriers to foraging behaviour that force them onto roads or other unsuitable places. A viable population needs access to about 90hectares of connected land. Hedgehogs are listed as a species of Principal Importance under the NERC Act (2006) and is classed as 'vulnerable' in England (Mammal Society, 2020).	Mitigation will be required in terms of during construction and after. Appendix 1 provides suitable avoidance measures during construction work and Appendix 3 provides measures for post-construction.

5 ENHANCEMENTS

5.1 ECOLOGICAL ENHANCEMENT

Wildlife features will be included as follows:

2x Woodcrete nestbox with 28mm opening. The box/es will be positioned on a mature tree or nearby building, at a height of no less than 2.5m from ground level. The opening of the box/es will face away from the prevailing wind.

1x built-in swift brick. The brick/s will be positioned at the top of the buildings wall plate, at least 5m from ground level. There will be a clear flight path into the nest brick, and the area will remain unobstructed.

1x built-in bee brick. The brick/s will be installed within the outer wall of the building, at various heights. The bee brick is designed for solitary, non-aggressive bee species and is therefore suitable for use in a domestic setting.

6 SUMMARY

Planning permission will be sought for the construction of a dwelling in the garden of 124 Cheltenham Road.

Arbor Vitae were commissioned by Boughton Butler to undertake a Preliminary Ecological Appraisal in order to assess the impact of the development on habitats and protected species.

The application will be for a self-build dwelling and therefore is exempt from Biodiversity Net Gain.

The habitats on site mainly limited to 'urban' habitat types, such as introduced shrub, vegetated garden, and non-native garden hedges. Habitats losses will be minimal, but replacement planting of fruit trees and the provision of a wildlife-friendly landscaping scheme are recommended.

No protected species are suspected to be on-site, and the plans are unlikely to have any impact, provided that all recommended mitigation measures are adhered to.

Ecological enhancement measures will include the following features:

- 2x Woodcrete nestbox with 28mm opening.
- 1x built-in swift brick.
- 1x built-in bee brick.

7 REFERENCES

ARG UK (2010). ARG UK Advice Note 5: Great Crested Newt Habitat Suitability Index. Amphibian and Reptile Groups of the United Kingdom

Bat Conservation Trust (2018) Bats and artificial lighting in the UK. *Bats and the Built Environment series,* Guidance Note 08/18. Institution of Lighting Professionals.

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CIEEM (2018) Guidelines for Ecological Impact Assessment in the UK and Ireland: Terrestrial, Freshwater, Coastal and Marine. Chartered Institute of Ecology and Environmental Management, Winchester.

Collins, J. (2016) Bat Surveys for Professional Ecologists: Good Practice Guidelines (4th edn). The Bat Conservation Trust, London.

Cresswell, W. and Whitworth, R., 2004. An assessment of the efficiency of capture techniques and the value of different habitats for the great crested newt. Natural England Research Reports, p.36.

Hundt L (2012) Bat Surveys: Good Practice Guidelines, 2nd edition, Bat Conservation Trust.

Jehle, Robert. (2000). The terrestrial summer habitat of radio-tracked great crested newts (Triturus cristatus) and marbled newts (T. marmoratus). Herpetological Journal. 10. 137-142.

JNCC (2010) Handbook for Phase 1 habitat survey - a technique for environmental audit, ISBN 0 86139 636 7.

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People's Trust for Endangered Species (2019) Hedgehogs and development. British Hedgehog Preservation Society. Accessed 11/04/2023. Available at: https://www.hedgehogstreet.org/wp-content/uploads/2019/03/Hedgehogs-and-developers-ZR.pdf

8 APPENDICES

Appendix 1. Reasonable Avoidance Measures Method Statement

A set of Reasonable Avoidance Measures will be adopted on site during all pre-construction, during, and post-construction phases. The following applies:

1. The site owner/site manager will ensure that anyone (including sub-contractors) undertaking construction, demolition and landscaping (both creation and management), is made aware of the potential for the site to support protected species, where to expect them, their protected status and the procedure to follow in the unlikely event that protected species are discovered during works. A copy of this Precautionary Method of Working will be always kept on site and available for inspection.
2. Should any hedgehogs or other protected species (i.e. common toads, reptiles, birds) be discovered during construction or other works on site, which are likely to be affected by the development, works will cease immediately. The owner/ site manager will then seek the advice of a suitably qualified and experienced ecologist and works will only proceed in accordance with the advice they provide.
3. Any vegetation clearance (i.e. removal of trees, scrub, hedgerows) or works to buildings shall not commence until a careful check for nesting birds has been completed, particularly during the main breeding season (months of March to August). If bird nests are found, then works in the area must stop until the chicks have fledged the nest.
4. All clearance works (i.e. clearance of log piles, debris, rough grass etc.) will be undertaken when hedgehog are likely to be fully active i.e. during the April to September period.
5. Clearance of dry-stone walls, logs, brash, stones, rocks or piles of similar debris will be undertaken carefully and by hand.
6. Clearance of tall vegetation will be undertaken using a strimmer or brush cutter with all cuttings raked and removed the same day. Cutting will only be undertaken in a phased way which may either include: Cutting vegetation to a height of no less than 30mm, clearing no more than one third of the site in any one day or; Cutting vegetation over three consecutive days to a height of no less than 150mm at the first cut, 75mm at the second cut and 30mm at the third cut;
7. Following removal of tall vegetation, remaining vegetation will be maintained at a height of 30mm through regular mowing or strimming to discourage common reptiles and amphibians from returning.

8. Ground clearance of any remaining low vegetation (if required) and any ground works will only be undertaken following the works in point 5) above.
9. Any trenches left overnight will be covered or provided with ramps to prevent animals falling into the trenches and being trapped. Excavations left overnight should be checked prior to filling. Any open pipes left overnight will be covered.

Appendix 2. Wildlife Sensitive Lighting Plan

The following measures will be incorporated into site-wide lighting design:

- Hedgerows and key habitat features including mature trees on the site will not be illuminated in order to retain dark movement corridors for nocturnal wildlife.
- Any exterior security or decorative lights to be installed on the development site will be less than 3 m from the ground and fitted with hoods to direct the light below the horizontal plane, at an angle of less than seventy degrees from vertical, and shall not be fixed to, or directed at, bat boxes or gables or eaves.
- Security lighting will be set on motion sensors with short timers (<1 minute) and will be LED with a passive infrared trigger.
- Lighting must be less than 3 lux at ground level and there shall be no light splay exceeding 1 lux along buildings, eaves or roof or adjacent hedgerows or trees.
- External lights will be hooded and directed toward the ground to reduce upward light spill.
- A warm white spectrum will be adopted throughout the scheme to reduce blue light component (<2700Kelvin).
- Internal luminaires will be recessed where installed in proximity to windows to reduce glare and light spill. LED luminaires will be used internally where possible due to their sharp cut-off, lower intensity, and dimming capability.
- Luminaires will always be mounted horizontally with an upward light ratio of 0%.

Bat Conservation Trust (2023) Bats and artificial lighting at night. *Bats and the Built Environment series,* Guidance Note 08/18. Institution of Lighting Professionals.

Appendix 3. Hedgehog Highways

Any new boundary fences and walls will present solid barriers to wildlife, particularly hedgehogs. This is important because a third of hedgehogs have been lost in the last 20 years, and one major cause is barriers to foraging behaviour that force them onto roads or other unsuitable places. A viable population needs access to about 90hectares of connected land. Hedgehogs are listed as a species of Principal Importance under the NERC Act (2006) and is classed as 'vulnerable' in

England (Mammal Society, 2020). A 13 x13 cm gap should be provided at the base of barriers between gardens so that all garden space is accessible.

Hedgehog highways will be implemented throughout the site, especially within garden plots which are enclosed by close board fencing. Each garden will be fitted with at least one 13x13cm gap at the base of the fence with the final positioning to be agreed with the project ecologist. Each hole will be fitted with a 'Hedgehog Highway' sign to ensure they are not blocked up by residents.

The People's Trust for Endangered Species 'Hedgehog Street' initiative is an advocate for this method: https://www.hedgehogstreet.org/help-hedgehogs/link-your-garden/

Appendix 4. Pre-commencement breeding bird inspection

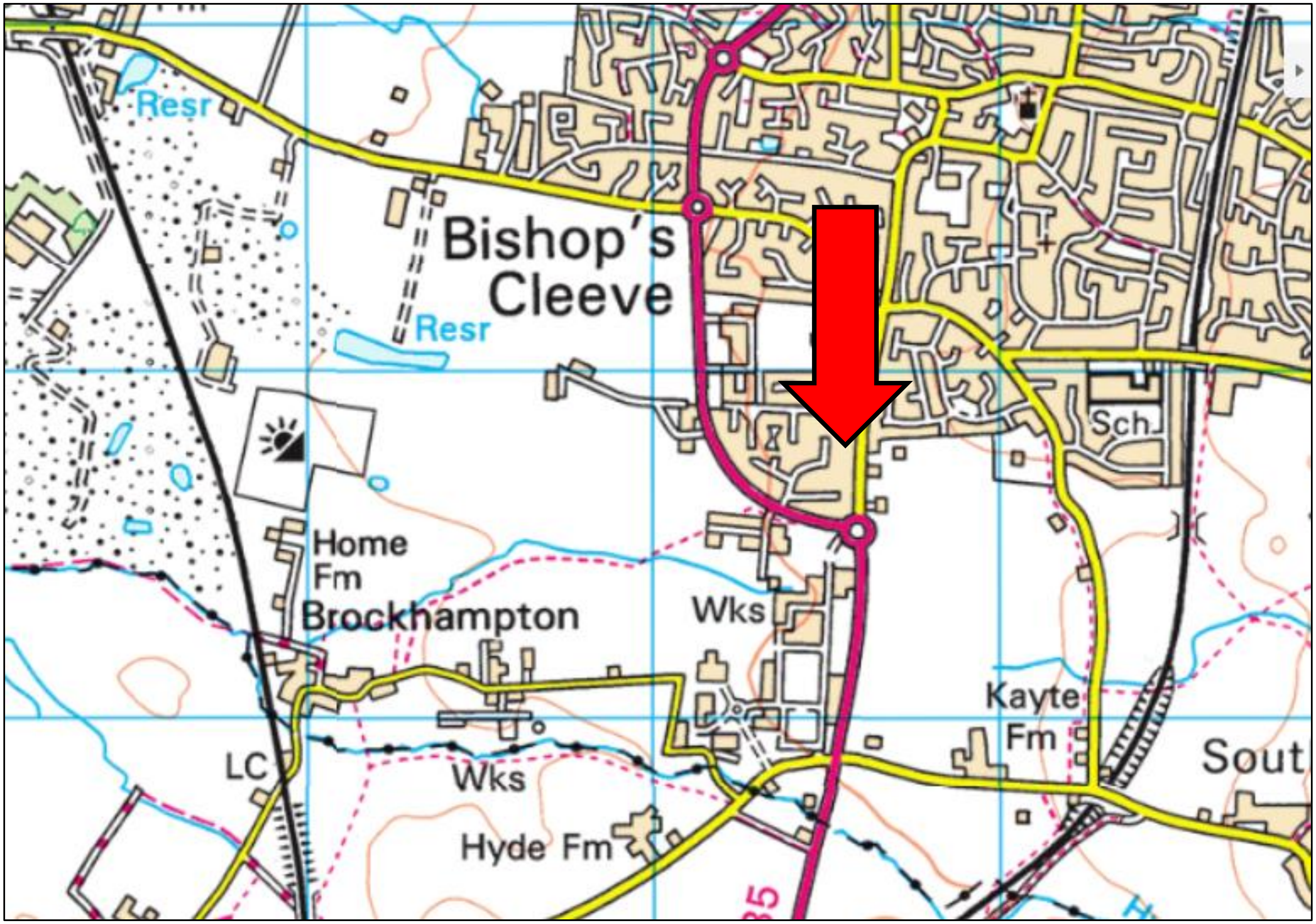
Prior to works starting on site, all structures or habitats which provide any suitable nesting habitat for breeding birds will be inspected thoroughly. If any sign of nesting behaviour is recorded, or nests are found, a minimum 5m buffer will be implemented around the nest until all young have fledged.

Appendix 5. Habitat Suitability Index assessment result

GCN HSI Calculator

	Pond Name	Pond 1
	Grid Ref	SO95512675
SI No	SI Description	
1	Geographic location	1
2	Pond area	0.05
3	Pond permanence	0.9
4	Water quality	0.67
5	Shade	1
6	Water fowl effect	0.33
7	Fish presence	0.33
8	Pond Density	0.2
9	Terrestrial habitat	0.33
10	Macropyhyte cover	0.3
HSI Score		**0.38**
Pond suitability (see below)		*Poor*

FIGURE 1 LOCATION



ARBOR VITAE

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FIGURE 2 AERIAL PHOTOGRAPH

ARBOR VITAE

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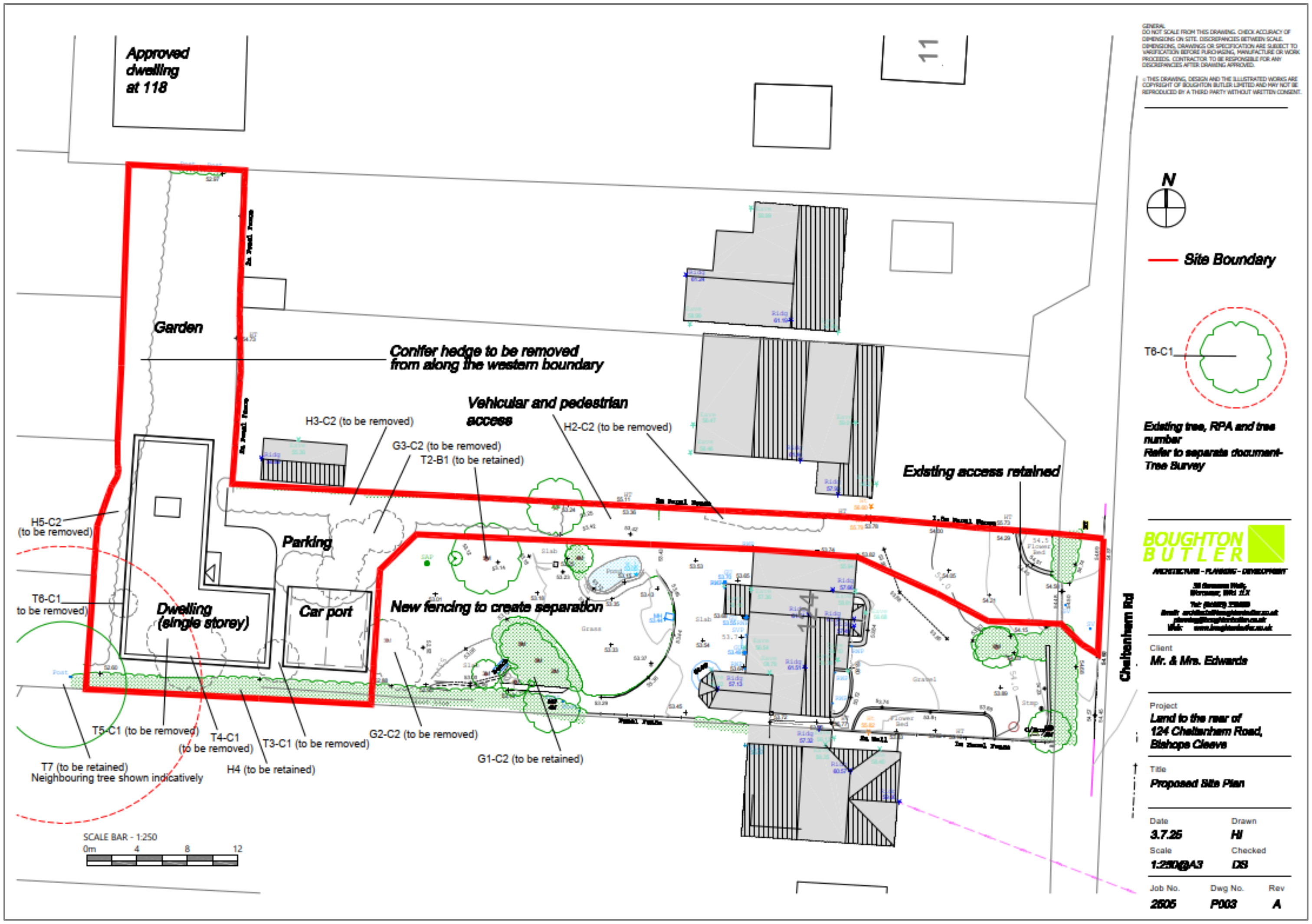
FIGURE 3 HABITAT MAP

FIGURE 4 PROPOSED SITE PLAN



APPENDIX 1 PHOTOGRAPHS

Exisitng driveway and access to be used.

Proposed route of new driveway.

Summer house and lawns.

Vegetable patch and H4 along west boundary.

G2 to be removed.

Existing house on site to be retained.

	
Ornamental fish pond to be retained	Existing access